

CHAPTER

37



Obesity

Learning Objectives

- ⇒ To know the clinical features of obesity
- ⇒ To perform the necessary physical examination of patient of obesity
- ⇒ To order and interpret the necessary laboratory investigations of obesity
- ⇒ To know the differential diagnosis of obesity

Introduction

Obesity is excess body fat accumulation that imparts health risk; a body weight of 20% excess over ideal weight, age, gender and height is considered as health risk.

Body Mass Index



Figure 160: Gradings of BMI

Etiology of obesity

This occurs when caloric input is exceeding the caloric utilization. In other words, chronic imbalance between energy intake and energy utilization, where in amount of energy intake exceeds amount of energy utilization.

- ⊗ Inadequate pushing of oneself away from the dining table causing overeating
- ⊗ Insufficient pushing of oneself out of the chair leading to inactivity and sedentary lifestyle
- ⊗ Genetic predisposition to develop obesity

- ⊗ Diets largely derived from carbohydrates and fats than protein rich diet
- ⊗ Secondary obesity which follows a number of underlying diseases such as hypothyroidism, Cushing's syndrome, insulinoma and hypothalamic disorders

Sequelae of obesity: Marked obesity is a serious health hazard, predispose to a number of clinical disorders.



Figure 161: Clinical features of obesity

⊗ **Pathological changes:** This is associated with increased adipose stores in the subcutaneous tissues, skeletal muscles, internal organs such as kidneys, heart, liver and omentum. Fatty liver is also most common in obese individuals.

⊗ **Metabolic changes:**

- **Hyperinsulinaemia:** Increase in insulin secretion is a feature of obesity. Many obese individuals exhibit hyperglycemia or frank diabetes despite hyperinsulinaemia. This is due to a state of insulin resistance consequent to tissue insensitivity.
- **Non-insulin dependent diabetes:** There is a strong association of non-insulin dependent diabetes mellitus with obesity. Obesity often exacerbates the diabetic state and in many cases weight reduction often leads to amelioration of diabetes.
- **Hypertension:** A strong association between hypertension and obesity is observed which is due to increased blood volume. Weight reduction leads to significant reduction in systolic blood pressure.
- **Hyperlipoproteinemia:** The plasma cholesterol circulates in the blood as low density lipoprotein (LDL) containing most of the circulating triglycerides. Obesity is strongly associated with VLDL and mildly with LDL. Total blood cholesterol levels are also elevated in obesity.
- **Atherosclerosis:** Obesity predisposes to development of atherosclerosis.
- **Coronary artery disease and stroke:** As a result of atherosclerosis and hypertension, there is increased risk of myocardial infarction and stroke in obese individuals.
- **Cholelithiasis:** There is 6 times higher incidence of gall stones in obese individuals, mainly due to increased total body cholesterol.

➤ **Osteoarthritis:** These individuals are more prone to develop degenerative joint disease due to wear and tear following trauma to joints as a result of large body weight.

➤ **Cancer:** certain cancers such as of endometrium and breast seem to be related to obesity. Particularly implicated are the diets derived from animal fats and meats. However, definite evidence in support of the role of diet in cancer is lacking.

Complications of obesity

- ⊗ Metabolic syndrome
- ⊗ Diabetes mellitus
- ⊗ Cardiovascular disease
- ⊗ Nonalcoholic hepatitis
- ⊗ Gall bladder disease
- ⊗ Gastroesophageal reflux disease
- ⊗ Obstructive sleep apnoea
- ⊗ Reproductive system diseases
- ⊗ Cancer
- ⊗ Osteoarthritis
- ⊗ Social and psychological problems

Diagnosis is done by analysis of BMI and waist circumference. BMI is calculated by the formula $W \text{ (kg)} / h^2 \text{ (m)}$. The normal BMI ranges between 18 to 25. If the BMI is more than 25, then it is indicative of overweight and if BMI crosses 30, then the diagnosis of obesity is made and if the BMI crosses 40, then hyperobesity is diagnosed. As far as waist circumference is concerned, Asian Indian men will have waist circumference $> 78 \text{ cm}$ and Asian Indian women will have waist circumference of $> 72 \text{ cm}$ may indicate obesity.

Key summary of the chapter

- ⊗ Obesity is excess body fat accumulation that imparts health risk; a body weight of 20% excess over ideal weight, age, gender and height is considered as health risk.
- ⊗ Etiology of obesity: This occurs when caloric input is exceeding the caloric utilization. In other words, chronic imbalance between energy intake and energy utilization, where in amount of energy intake exceeds amount of energy utilization.
- ⊗ Complications of obesity are as follow: metabolic syndrome, diabetes mellitus, cardiovascular disease, nonalcoholic hepatitis, gall bladder disease, gastro oesophageal reflux disease, obstructive sleep apnoea, reproductive system diseases, cancer, osteoarthritis, social and psychological disturbances.
- ⊗ Diagnosis is done by analysis of BMI and waist circumference.

Self-Assessment: Questions for Review

Short essay questions

- ⊗ Describe obesity
- ⊗ Define obesity, write the causes and clinical features and complications of obesity.
- ⊗ Describe how to make the diagnosis of obesity

MCQs

1. BMI is being calculated by the formula
 - a. $W \text{ (gm)} / h^2 \text{ (cms)}$;
 - b. $W \text{ (kg)} / h^2 \text{ (cms)}$;

Chapter resources

- c. W^2 (Kg)/h (m);
- d. W (kg)/h² (m)

Answer: (d)

2. The term BMI in the context of diagnosis of obesity refers to

- a. Body Mass Index;
- b. Bone Mass Index;
- c. Body Muscle Index;
- d. Bone Marrow Index

Answer: (a)

3. The diagnosis of obesity is made when the BMI crosses

- a. 25; b. 40;
- c. 30; d. 18

Answer: (c)

4. The normal range of BMI in human being is

- a. 18 to 25; b. 15 to 20;
- c. 25 to 30; d. 30 to 40

Answer: (a)

5. Mr. X is visiting one obesity clinic for normal routine checkup. The doctor calculates his BMI as 28. As per the assessment of BMI calculation formula, Mr. X is having

- a. Undernutrition; b. Obesity;
- c. Overweight; d. Hyperobesity

Answer: (c)

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